



Marine Biology

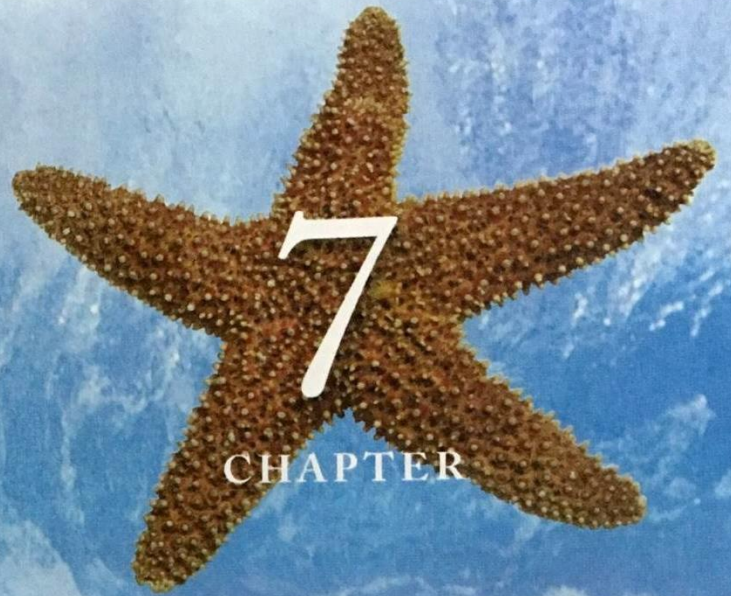
海洋生物学



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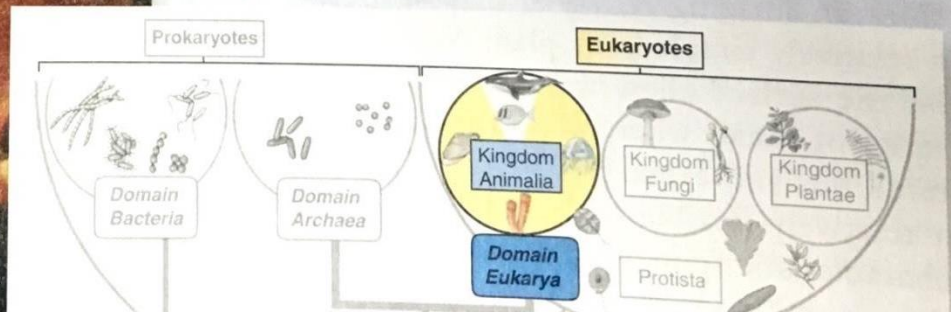
Ocean College, Zhejiang University



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CHAPTER

Marine Animals Without a Backbone



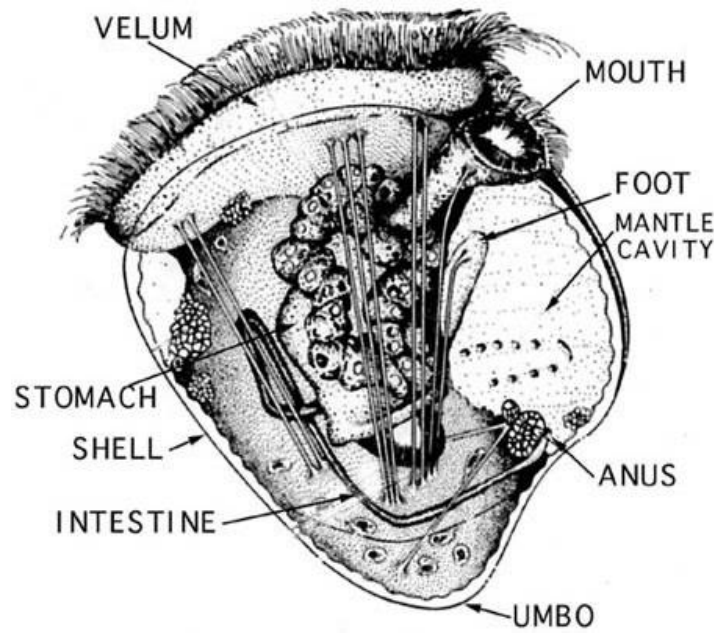
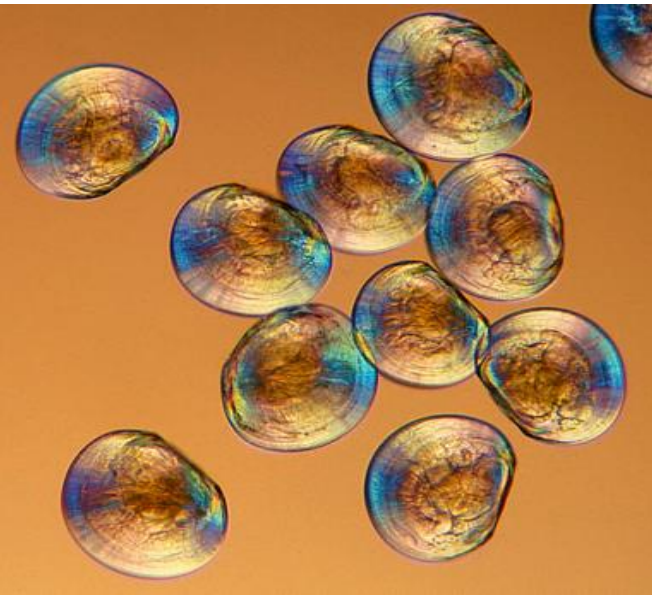
• Major Phyla of Marine Invertebrates

- | | |
|-----------------------|----------------------------|
| 1. Sponges (海绵动物) | 10. Molluscs (软体动物) |
| 2. Cnidarians (腔肠动物) | 11. Bryozoans (苔藓虫) |
| 3. Flatworms (扁形虫) | 12. Phoronids (帚虫) |
| 4. Ribbon worms (纽虫) | 13. Brachiopods (腕足类) |
| 5. Nematodes (线虫) | 14. Arrow worms (箭虫) |
| 6. Annelids (环节动物) | 15. Echinoderms (棘皮动物) |
| 7. Sipunculans (星虫动物) | 16. Hemichordates (半索海生动物) |
| 8. Echiurans (螭虫动物) | 17. Tunicates (被囊动物) |
| 9. Arthropods (节肢动物) | 18. Lancelets (文昌鱼) |

11. The Molluscs (Mollusks)

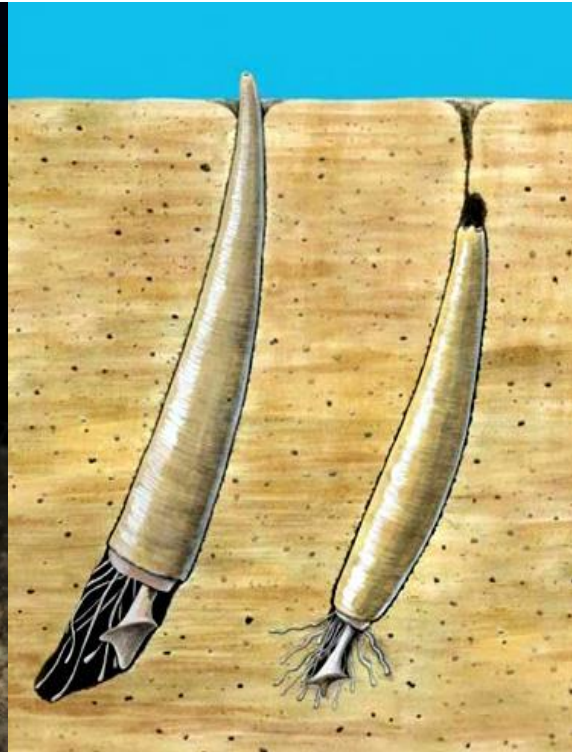
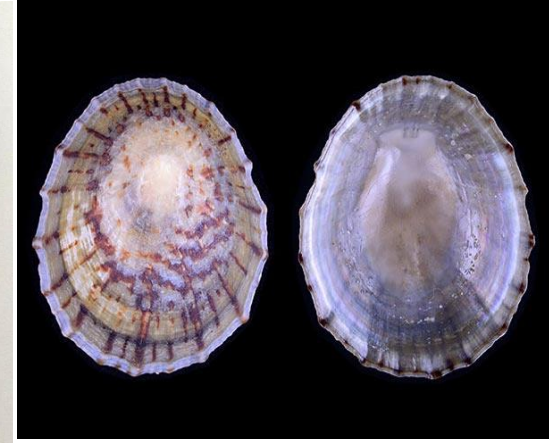
- Phylum Mollusca
- Over 200,000 species, second-largest animal phylum
- Name means "**soft body**", bilateral symmetry
- Basic body plan:
 - **Head, muscular foot** and **visceral (内脏的) mass** in most species
 - **Mantle**: secretes shell, waste disposal, sensory reception, reproduction
 - Many have a **shell of calcium carbonate**
 - **Radula (齿舌)** for grazing is unique to this group

- Can be deposit feeders, filter feeder, carnivores, or using radula for scraping algae, encrusting animals off substrates
- **Well developed nervous system**
- **Open circulatory system & complete digestive system**
- Trochophore larvae develops into a **planktonic veliger larvae (缘膜幼体/面盘幼虫)** complete with shell (miniature version of adult)

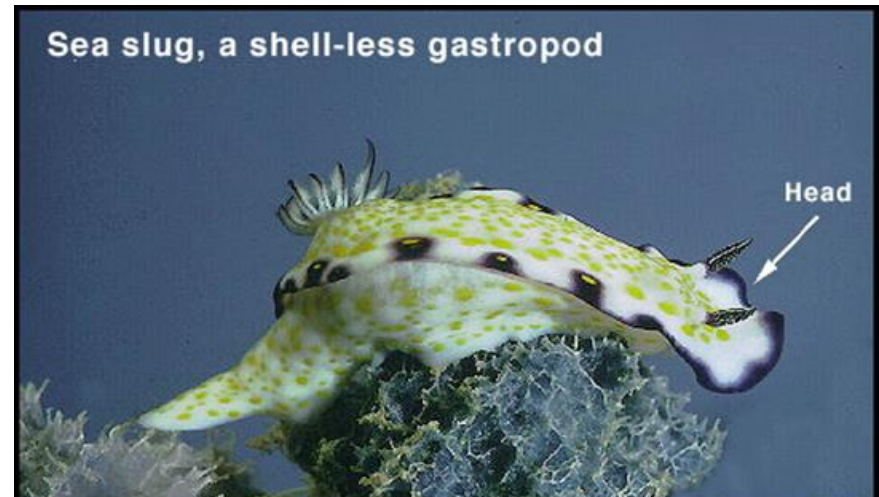


Class including:

- Aplacophora: **无板纲**
- Monoplacophora: **单板纲**
- Polyplacophora: **多板纲**
- Bivalve: **双壳纲**
- Gastropoda: **腹足纲**
- Scaphopoda: **掘足纲**
- Cephalopoda: **头足纲**

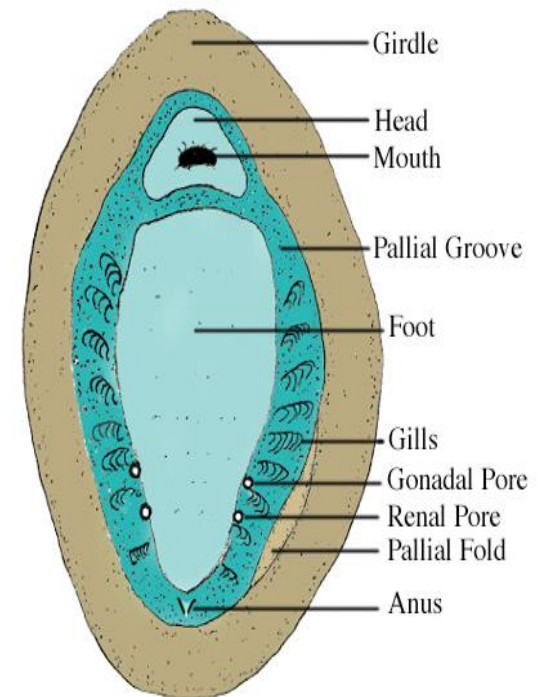
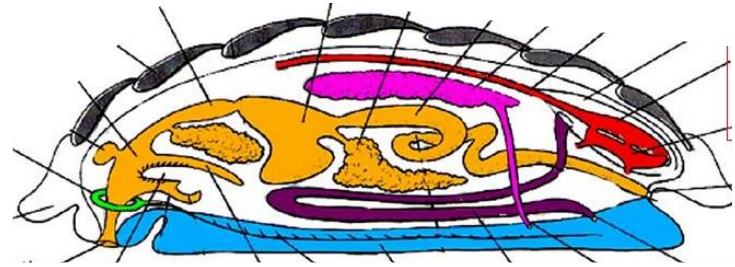


- Polyplacophora (Chiton), Gastropods (snails, sea slug); Bivalves (oysters, clams, mussels); Cephalopods (octopus, cuttlefish)



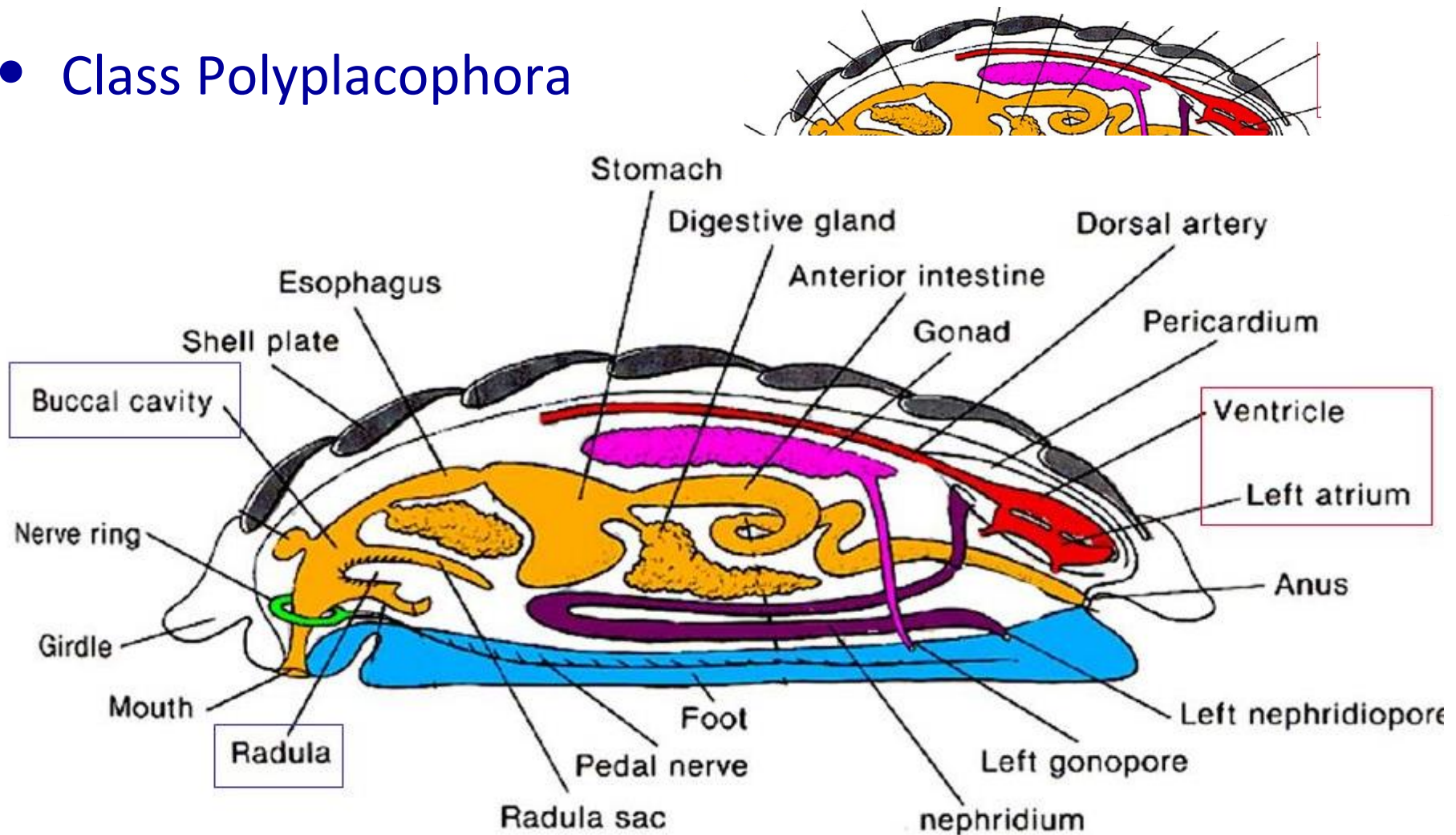
11.1 Chitons (石鳖)

- Class Polyplacophora
- 800 species, **all marine**
- Dorsal shell of 8 plates
- Ventral muscular foot
- Ventral mouth with radula
- Mostly found in shallow water, coastal environments of hard substrate
- Many graze on algae & small animals in marine intertidal zone (area between high and low tides)

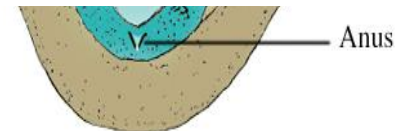


11.1 Chitons (石鳖)

- Class Polyplacophora



in marine intertidal zone (area between high and low tides)



11.2 Bivalves

- Clams, Oysters, Scallops, Mussels, etc.
- Two shells or “valves”
- Oldest part of the shell is called the **umbo** (壳嘴)
- Shell grows out from the umbo in **concentric rings**
- **No head present**
- **Adductor muscles** secure valves together



Class

Bivalve



Subclass

- Cryptodonta (隐齿亚纲)

e.g. 蛭螂目 (我国不产)



- Palaeotaxodonta (古多齿亚纲)

e.g. 胡桃蛤目



- Pterimorphia (翼形亚纲)

e.g. 蚌目、贻贝目、珍珠贝目、牡蛎目



- Palaeoheterodonta (古异齿亚纲)

e.g. 蚌目



- Heterodonta (异齿亚纲)

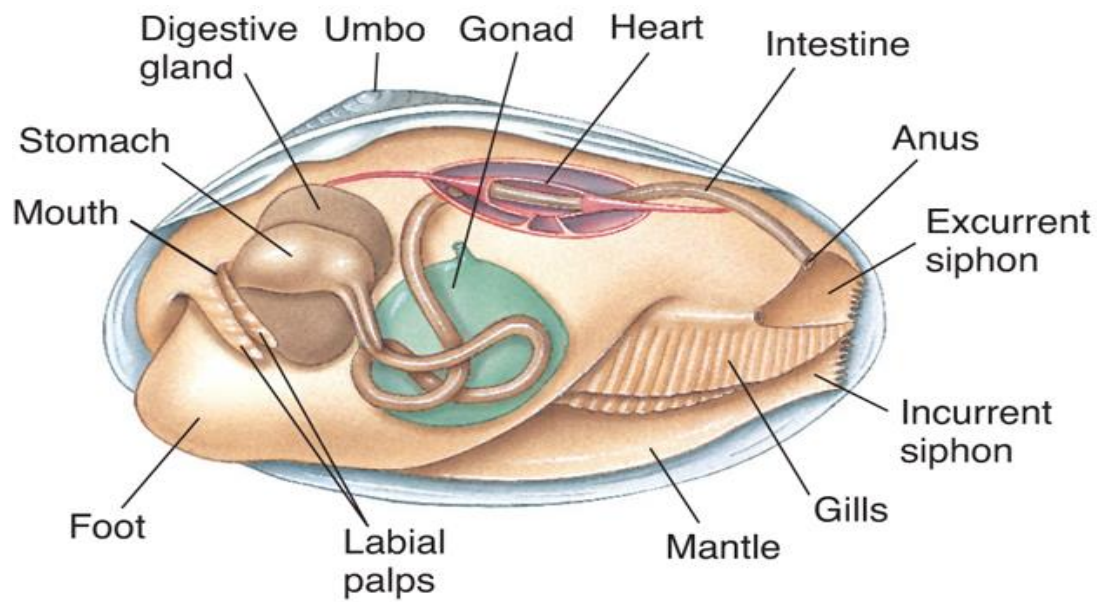
e.g. 帘蛤目、海螂目



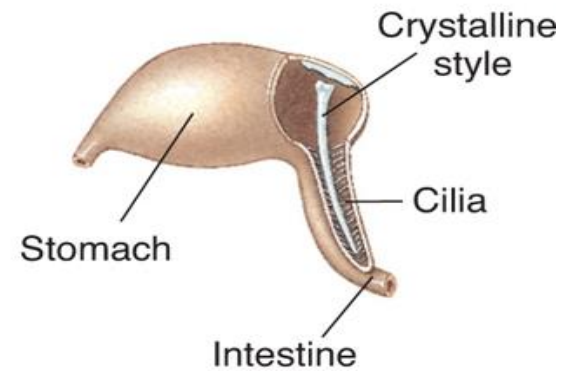
- Anomalodesmacea (异韧带亚纲)

e.g. 笋螂目、隔鳃目

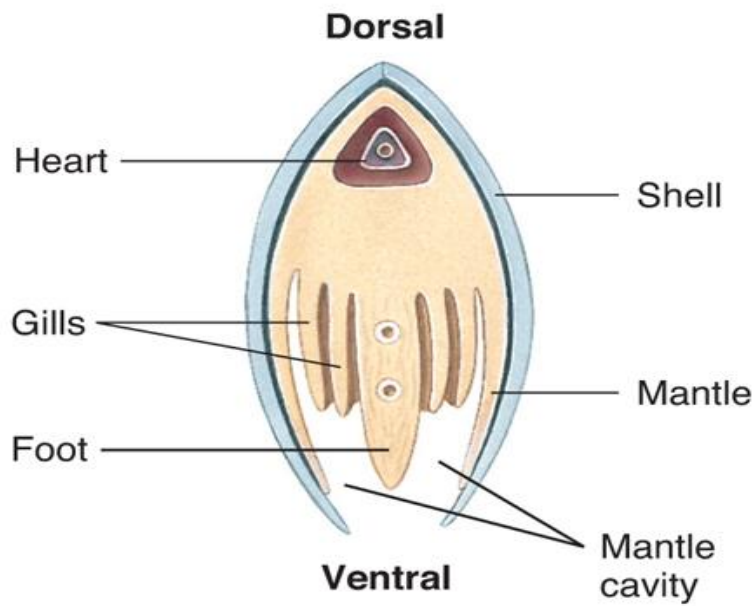




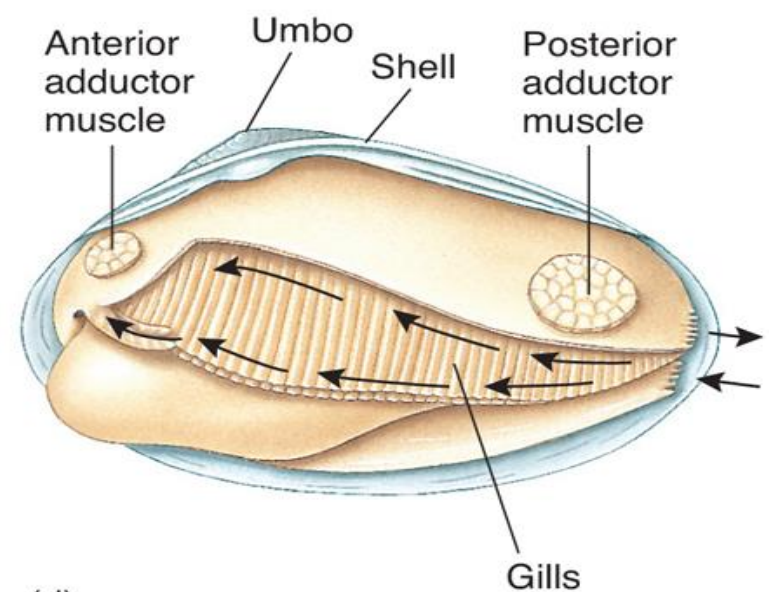
(a)



(b)



(c)



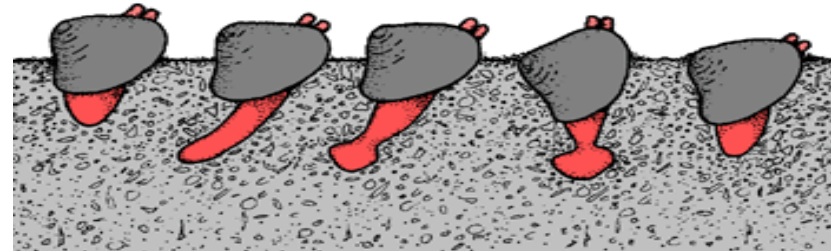
(d)

- **Muscular foot** used for burrowing in bottom and other locomotion (移动), the size varies among marine bivalves

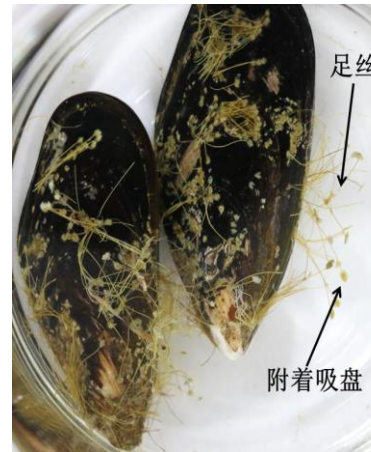
- **Clams** have a **big hatchet-shaped foot** for moving around and for burrowing in mud or sand



- **Oyster and mussels:** **small foot** because these animals attach themselves to hard objects via **byssus threads (足丝)**, or grow attached to each other

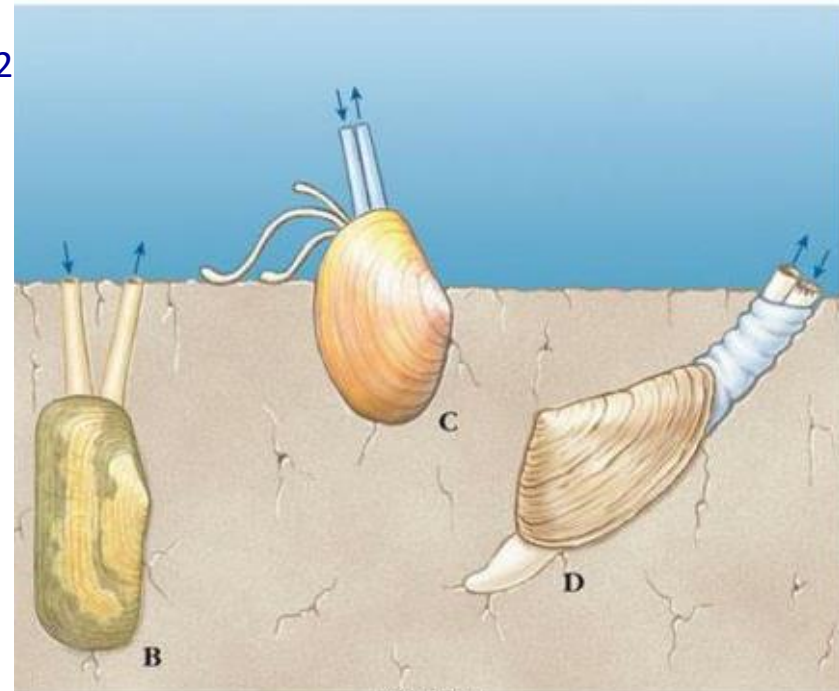
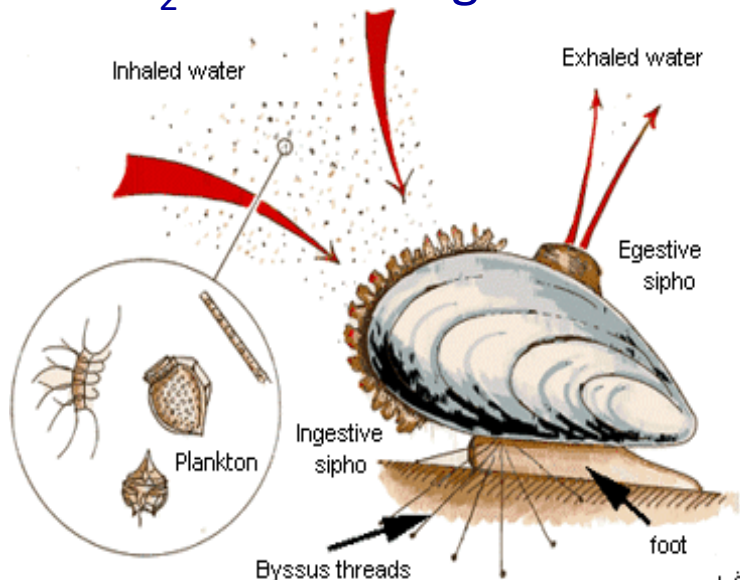


- **Scallops** don't use their **small foot** to move around either. They swim by **jet propulsion**, clapping their shells together and forcing water out the rim (边缘)



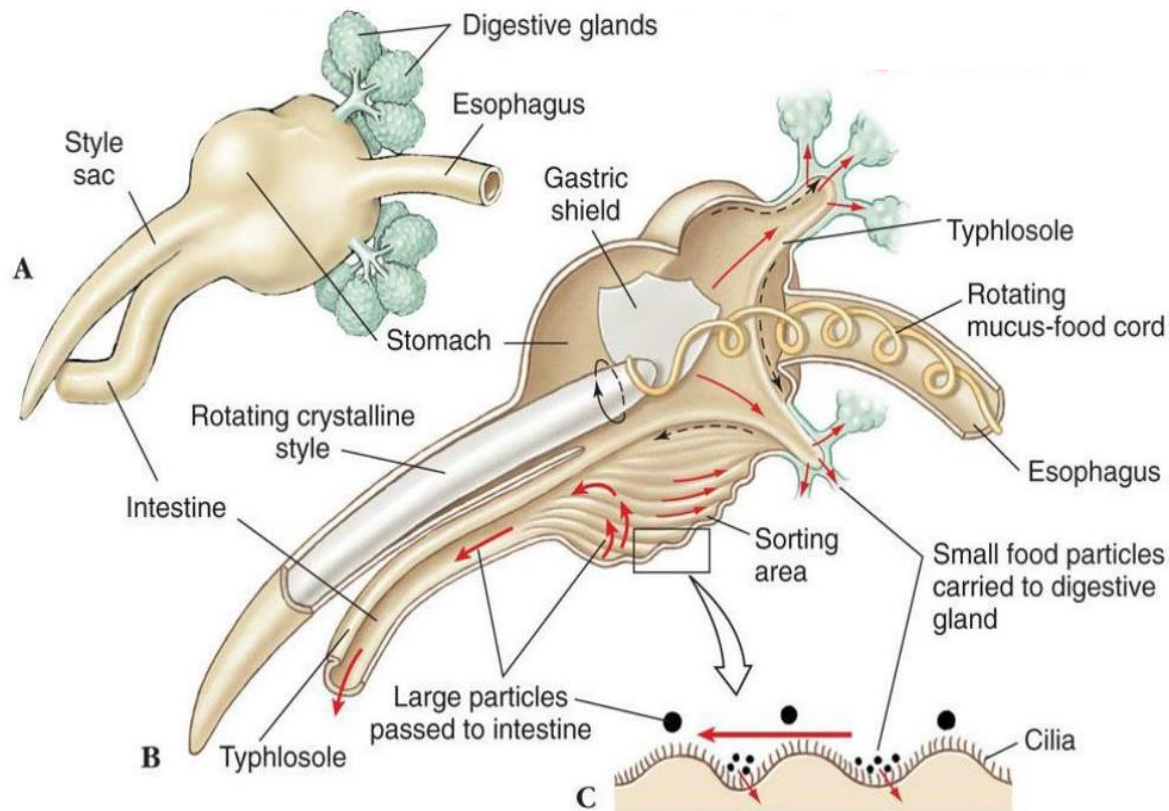
• Siphons (虹吸管)

- Water enters and leaves a bivalve by way of two tubes: **siphons**. One takes in water while the other expels water and wastes
- Water taken in contains oxygen and food particles consisting of detritus and plankton
- As the water flows across the gills, O_2 and CO_2 are exchanged

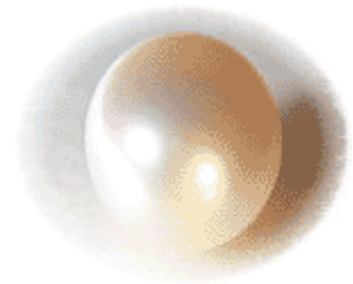
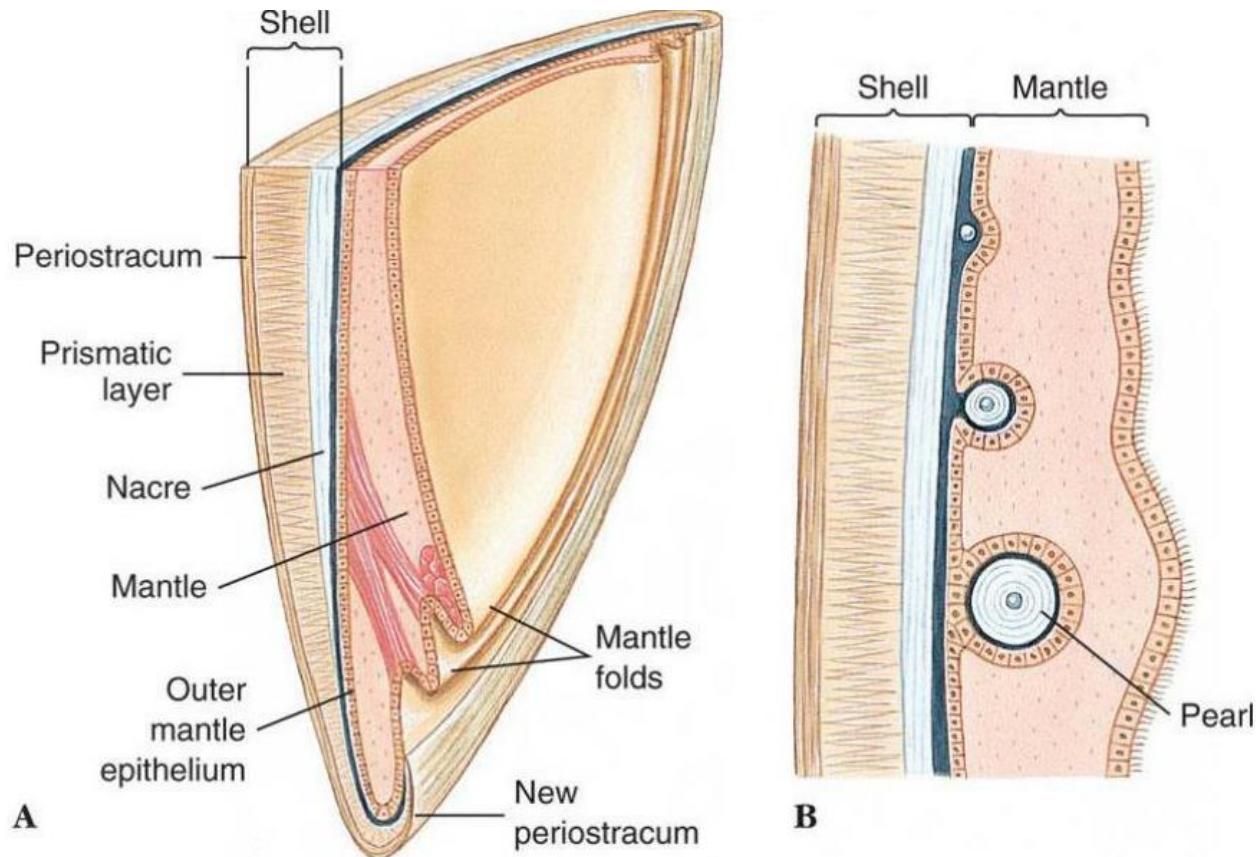


A: © R. Harbo

- **Mucus** on the **gills** traps microscopic food particles, tiny hair-like cilia on the gills move the food-laden mucus toward the mouth
- **Liplike structures** called **palps** help sort the food and direct it into the mouth
- **Bivalves do not have a radula.** The food suspended in mucus moves through the digestive organs, which break it down and absorb it



- In some bivalves, such as oysters and mussels, **mantle tissue secretes a pearly substance** that coats any irritating particles (e.g.: bits of gravel), that lodge between the mantle and the shell
- As coats of this substance build up they form a pearl





- Bivalves as food

- Bivalves are valuable as food
- Except for the shell, bivalves can be eaten whole

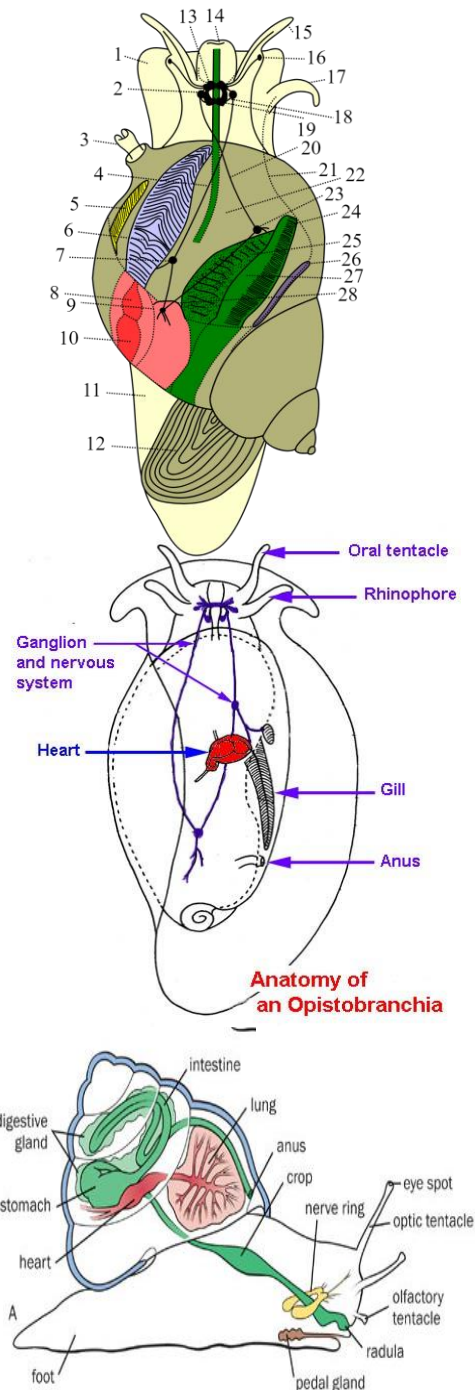


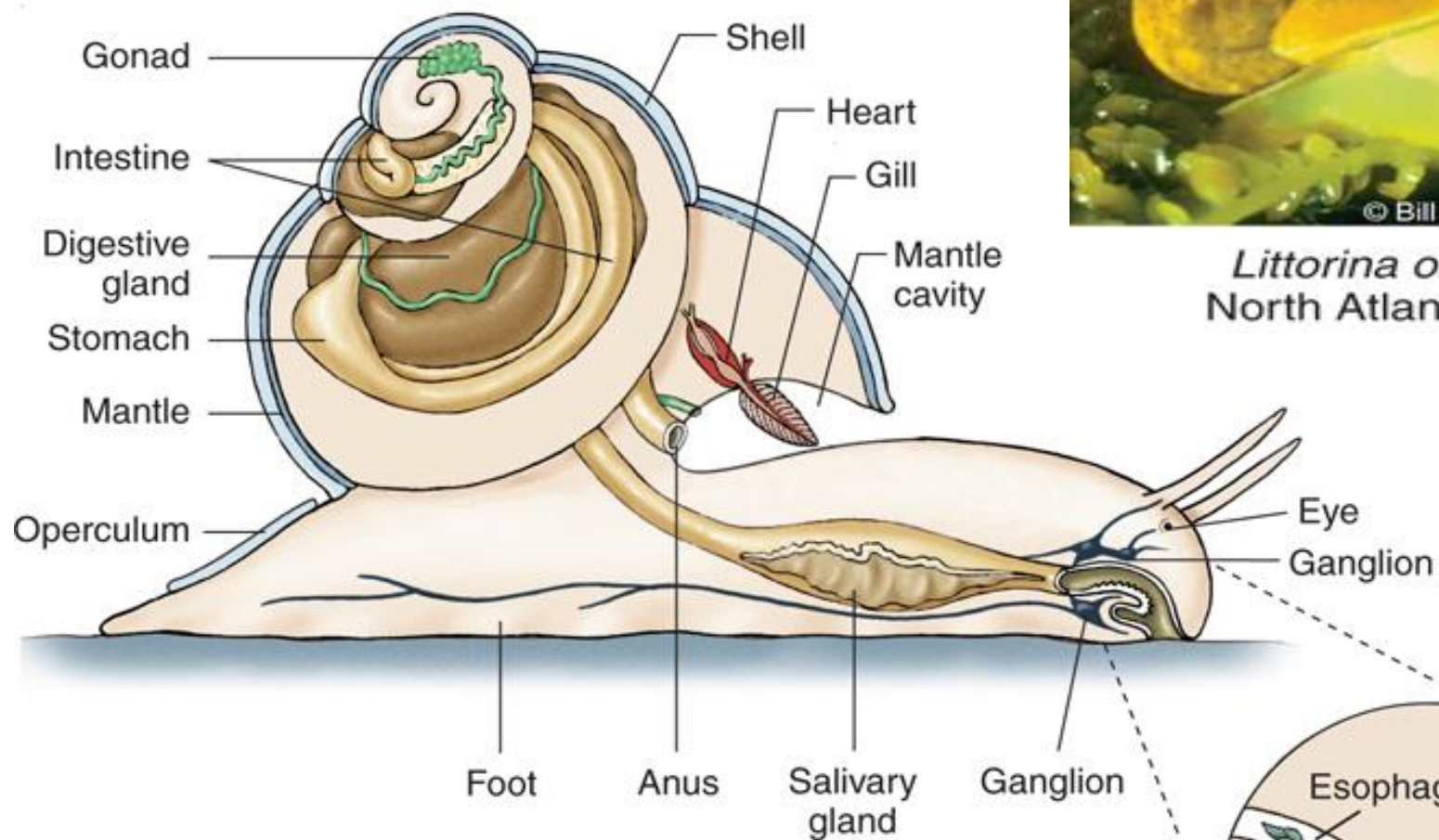
2014-2019年中国扇贝海水养殖产量及增速



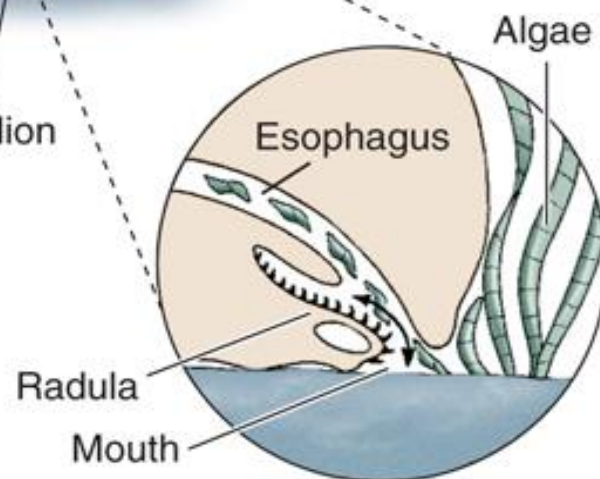
11.3 Gastropods

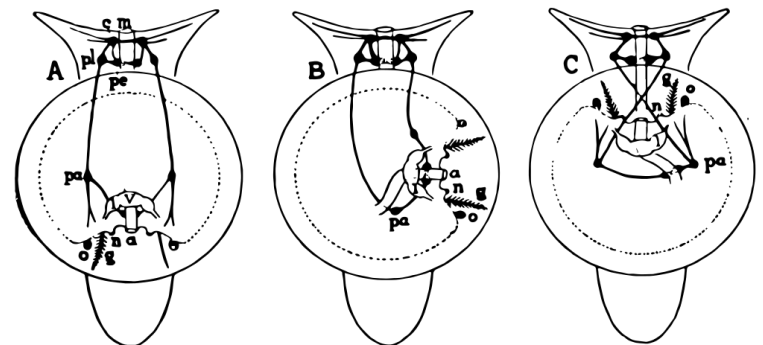
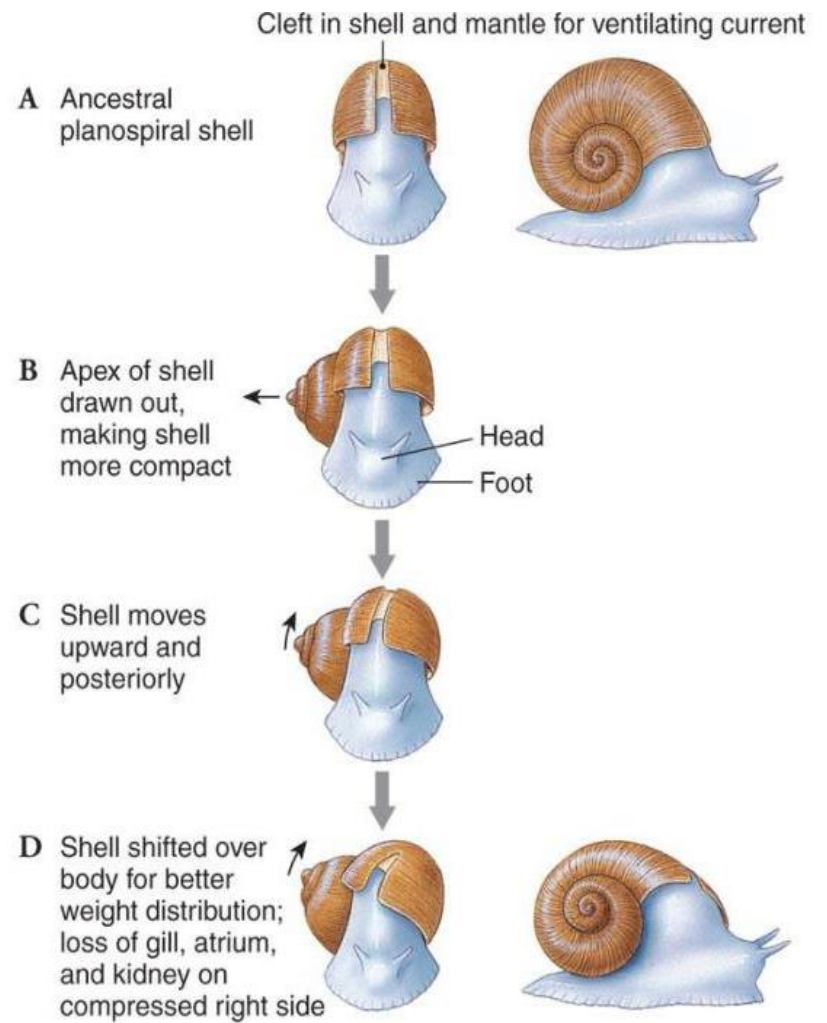
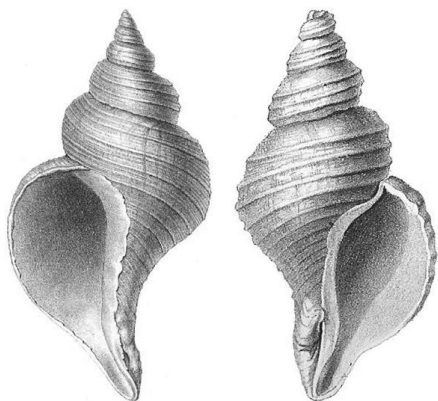
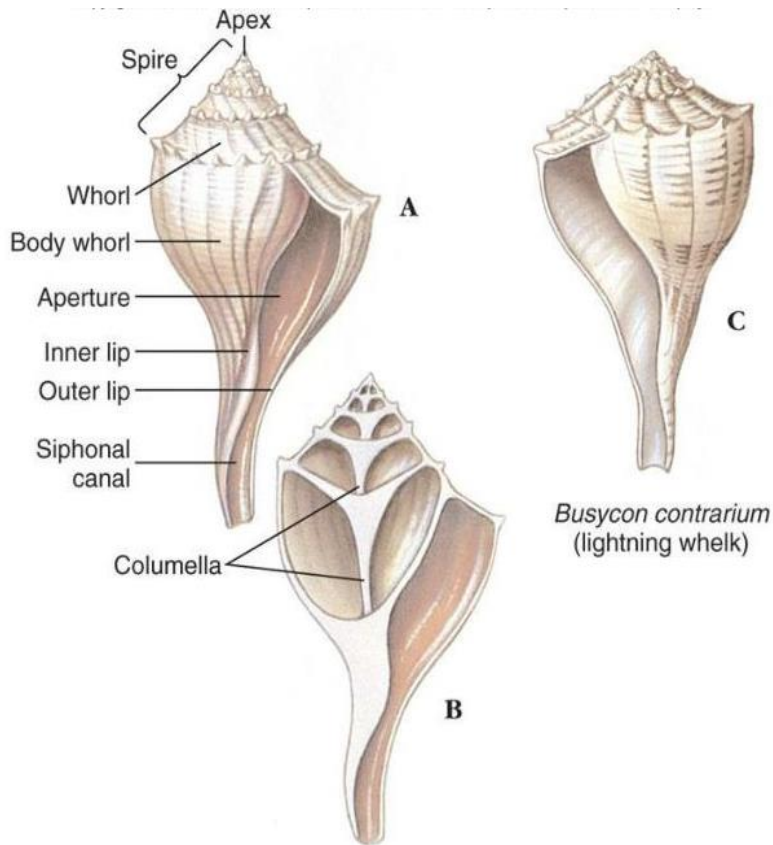
- **Largest class of molluscs**, about 75,000 species
- Name means "**belly-footed**"
- **Coiled shell** on most species
- No shell on sea slugs (nudibranchs, 裸鳃亚目)
- **Radula** for grazing on plants in most, as well as prey capture
- Some are deposit feeders, some are carnivorous, some will even prey on members of the same species)
- Including Prosobranchia (前鳃亚纲), Opisthobranchia (后鳃亚纲), Pulmonata (肺螺亚纲)



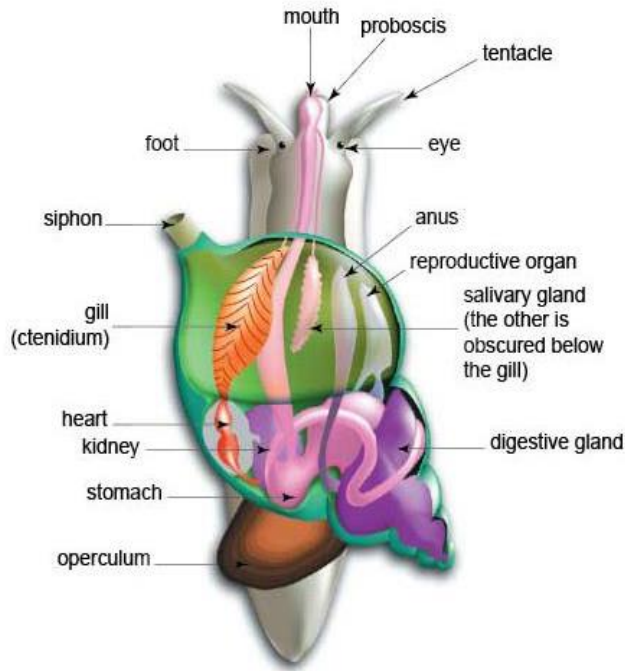


Littorina obtusata,
North Atlantic Ocean

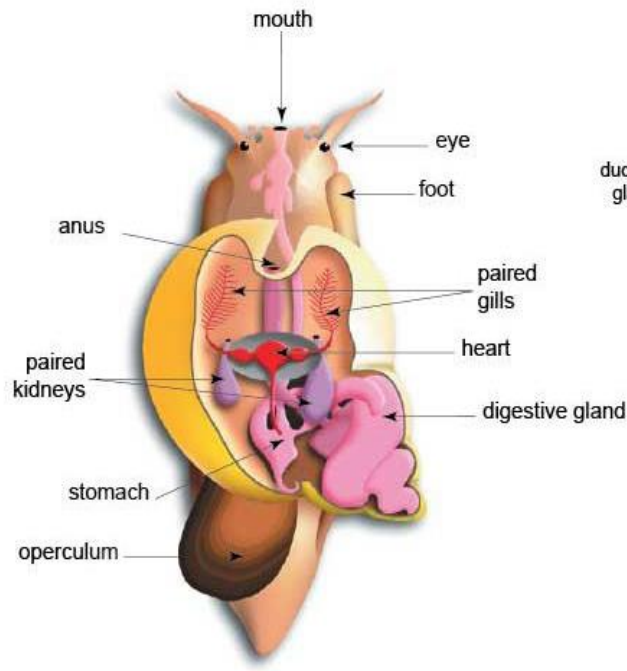




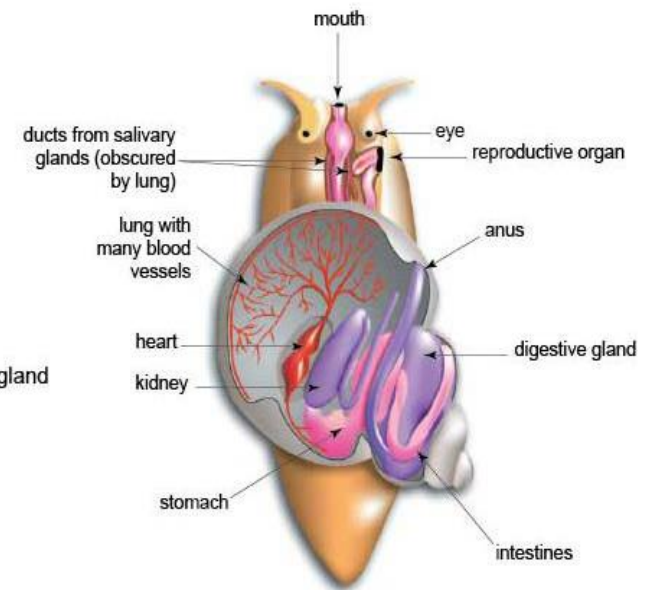
- Gill of gastropod



Caenogastropod



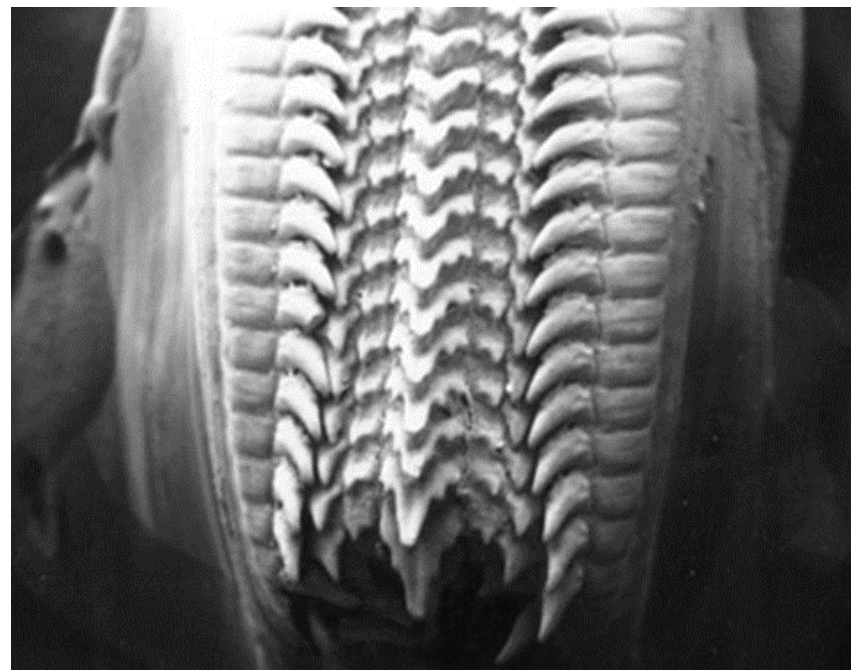
Archaeogastropod



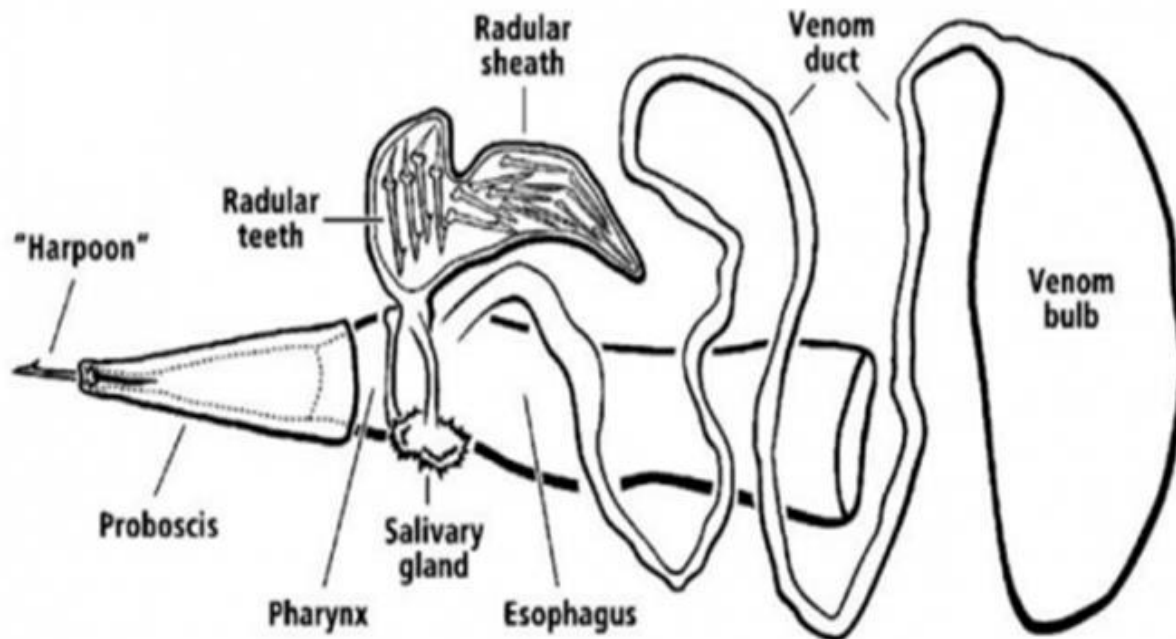
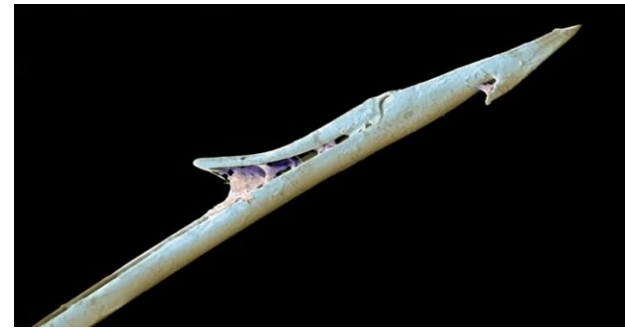
Pulmonate gastropod

- Eating behavior

- ❑ Radula: rasp (锉刀) like tongue in the mollusks' mouth
- ❑ Herbivorous (plant-eating/vegetarian) snails have a mouth with radula containing many rows, each row with five to seven complex teeth
- ❑ The snail uses its radula like a rasp, scraping off small bits of food



- ❑ Some snails, e.g. Cone (圆锥体) shells, have barbed radula tooth fires through the proboscis, an extension of the mouth
- ❑ Can produce a poison strong enough to kill humans who handle them carelessly; their poison is a neurotoxin, one that attacks and destroys nerves





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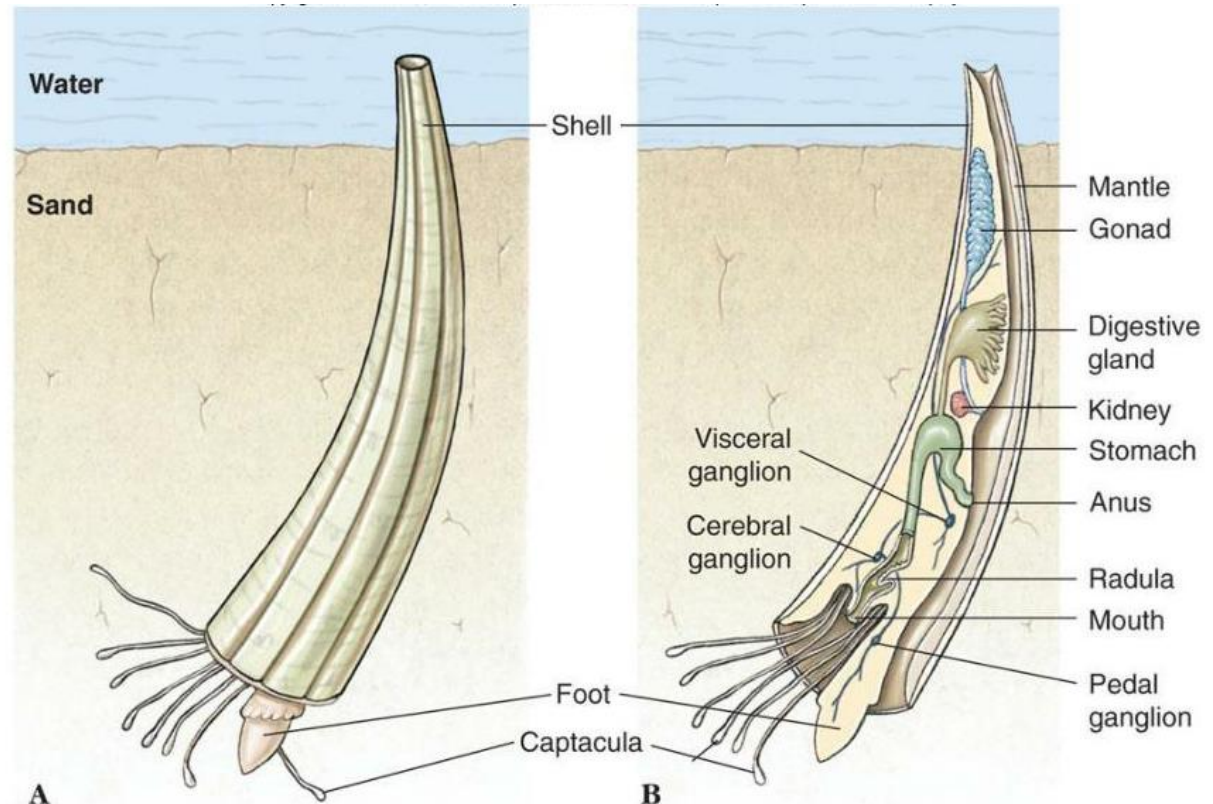
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11.3 Scaphopoda

- ▣ No head, lack gills (mantle absorbs oxygen from water)
- ▣ Ciliated, contractile tentacles (captacula, extending from foot) surround mouth to find food in sediment and carry it to the mouth
- ▣ No eyes or other distinct sensory organs



- Homework

